

تمارين رياضيات //1// مشترك

(I) Find The Domain and The Range of each Function

التابع Function	مجموعة التعريف D(f) Domain (x)	مجموعة القيم R(f) Range (y)	التابع Function	مجموعة التعريف D(f) Domain (x)	مجموعة القيم R(f) Range (y)
$f(x) = x^2$			$f(x) = \frac{1}{x}$		
$f(x) = \sqrt{x}$			$f(x) = \sqrt{4-x}$		
$f(x) = \sqrt{1-x^2}$			$f(x) = x $		
$f(x) = \sin(x)$			$f(x) = \cos(x)$		
$f(x) = \ln(x)$			$f(x) = \ln(-x)$		
$f(x) = e^x$			$f(x) = e^{-x}$		

(II) Graph The Functions

$f(x) = \begin{cases} x^2, & x \geq 0 \\ -x, & x < 0 \end{cases}$	$f(x) = \begin{cases} \sqrt{x}, & x \geq 0 \\ \sqrt{-x}, & x < 0 \end{cases}$
$f(x) = \begin{cases} e^x, & x < 0 \\ e^{-x}, & x \geq 0 \end{cases}$	$f(x) = \begin{cases} e^x, & x \leq 0 \\ \ln(x), & x > 0 \end{cases}$
$f(x) = \begin{cases} -x, & x < 0 \\ x^2, & 0 \leq x \leq 2 \\ 3, & x = 2 \end{cases}$	$f(x) = \begin{cases} 0, & x < 0 \\ 2, & 0 \leq x < 2 \\ 5, & 2 \leq x < 3 \\ 7, & 3 \leq x < 4 \\ 5, & x \geq 4 \end{cases}$
$f(x) = \begin{cases} 2, & 0 \leq x < 1 \\ 0, & 1 \leq x < 2 \\ 2, & 2 \leq x < 3 \\ 0, & 3 \leq x \leq 4 \end{cases}$	$f(x) = \begin{cases} A, & 0 \leq x < \frac{T}{2} \\ -A, & \frac{T}{2} \leq x < T \\ T, & 2 \leq x < \frac{3T}{2} \\ -A, & \frac{3T}{2} \leq x \leq 2T \end{cases}$ حيث A, T ثابتان عدديان
$f(x) = \begin{cases} 4-x^2, & x \leq 1 \\ x^2+2x, & x > 1 \end{cases}$	$f(x) = \begin{cases} \frac{1}{x}, & x < 0 \\ x, & x \geq 0 \end{cases}$
$f(x) = \begin{cases} x^2, & -\infty < x < 0 \\ 2x, & 0 \leq x < 1 \\ 2, & 1 \leq x < \infty \end{cases}$	$f(x) = \begin{cases} 1, & -2 \leq x < 0 \\ 0, & 0 \leq x < 2 \\ -1, & 2 \leq x < 4 \end{cases}$
$f(x) = \begin{cases} 1-x, & 0 \leq x < 1 \\ 2-x, & 1 \leq x < 2 \end{cases}$	

(III) Graph each function by starting with the graph of the standard functions and applying an appropriate transformation

$f(x) = x + 2 - 1$	$f(x) = (x + 1)^2 + 1$
$f(x) = x - 1$	$f(x) = x + 1$
$f(x) = (x + 1)^2$	$f(x) = x - 2 + 3$
$f(x) = (x + 3)^3 - 2$	$f(x) = x + 2 $
$f(x) = e^x + 1$	$f(x) = \sin(x) + 1$
$f(x) = \cos(x) + 1$	$f(x) = x^2 + 2$
$f(x) = (x - 2)^2 - 1$	$f(x) = x + 1 + 3$

(IV) Composite Function

Find $(f \circ g)(x)$, $(g \circ f)(x)$, $(f \circ g)(x)$, ..., $(g \circ f)(1)$ for following Functions.

$$\begin{aligned} f(x) &= x^2 + 1 & \text{and} & & g(x) &= 2x \\ f(x) &= 2x^2 + x & \text{and} & & g(x) &= \sqrt{x^2 + 1} \\ f(x) &= \ln(x) & \text{and} & & g(x) &= e^x \\ f(x) &= x^2 + 2x & \text{and} & & g(x) &= \sqrt{x + 1} \\ f(x) &= \ln(x^2 + 2) & \text{and} & & g(x) &= e^x \end{aligned}$$

If $f(x) = x/(x + 3)$ find $g(x)$ as that $(f \circ g)(x) = x$

If $f(x) = 2x^3 - 4$ find $g(x)$ as that $(f \circ g)(x) = x + 2$

If $f(x) = e^{x-3}$ find $g(x)$ as that $(f \circ g)(x) = 3x^2 + 1$

If $f(x) = \sqrt{x + 2}$ find $g(x)$ as that $(f \circ g)(x) = x$

(V) Find the Domain , Range, Graph, $f^{-1}(x)$, Domain $f^{-1}(x)$ and show that $(f \circ f^{-1})(x) = x$ for following function:

$$f(x) = \frac{2x + 3}{x - 1} \quad D(f) = \mathbb{R} \setminus \{1\}, \quad R(f) = \mathbb{R} \setminus \{2\}$$

$$f(x) = \frac{2x + 2}{1 - 2x}$$

$$f(x) = \frac{3x + 1}{2 - x}$$

$$f(x) = \frac{3x + 1}{x + 2}$$

$$f(x) = \frac{ax+b}{a'x+b'} \Rightarrow R(f) = \mathbb{R} \setminus \left\{ \frac{a}{a'} \right\}$$

- ملاحظة:

(VI) Solve for the angle θ where $0 \leq \theta \leq 2\pi$:

$\cos^2(\theta) = \sin^2(\theta)$	$\sin(2\theta) + \cos(\theta) = 0$
$\cos(2\theta) + \cos(\theta) = 0$	$\cos^2(2\theta) = \frac{1}{4}$

(VII) Solve for y in terms of x as appropriate

$\ln(y^2 - 9) + \ln(x + 3) = \ln(y - 3) + \ln(x^2)$	$e^{4y}e^{2x} = e^{x^2}e^3$
$\ln(y^2 - 9) - \ln(y - 3) = \ln(x^2) - \ln(x + 3)$	$e^y = 3x^2 + 2$
$\ln(y^2 - 16) + \ln(x + 2) = \ln(y + 4) + \ln(x)$	$\ln y + 4 = 3x + 1$
$\ln(y^2 - 1) - \ln(y + 4) = \ln(\sin(x))$	$e^{\sqrt{y}} = e^{\ln(x^2)}$
$e^y = e^{x^2}e^{3x+2}$	

(VIII) Find the inverse functions of the following functions:

$f(x) = x + 1$	$f(x) = \frac{1}{2}x - 2$
$f(x) = e^x$	$f(x) = \ln(x)$

تمارين رياضيات //1// الجزء الثاني

I- Find the limits:

أوجد النهايات:

$\lim_{x \rightarrow 2} \frac{\sqrt{x^2 + 12} - 4}{x - 2}$	$\lim_{x \rightarrow 0} \sin \left(\frac{\pi + \tan(x)}{\tan(x) - 2\sec(x)} \right)$	$\lim_{x \rightarrow 0} \frac{\sin(x)}{x}$
$\lim_{x \rightarrow \infty} [\sqrt{x^2 + 1} - \sqrt{x}]$	$\lim_{x \rightarrow 1} \sin^{-1} \left(\frac{x - 1}{x^2 - 1} \right)$	$\lim_{x \rightarrow 0} \frac{\tan(x)}{x}$
$\lim_{x \rightarrow 0^+} \frac{\sqrt{5} - \sqrt{x^2 + 4x + 5}}{x}$	$\lim_{x \rightarrow 1} \cos^{-1} \left(\frac{x - 1}{x^2 - 1} \right)$	$\lim_{x \rightarrow 0} \frac{1 - \cos(x)}{\sin(x)}$
$\lim_{x \rightarrow \infty} \frac{x^2 - 5x + 3}{8x^2 - 4}$	$\lim_{x \rightarrow \infty} \tan^{-1} \left(\frac{x^2 - 1}{x - 1} \right)$	
$\lim_{x \rightarrow 0} \tan \left(\frac{\pi}{4} \cos(\sin(x)) \right)$	$\lim_{x \rightarrow 0} \left(\frac{1}{\sin^{-1}(x)} - \frac{1}{\tan^{-1}(x)} \right)$	

II- Find the limits:

أوجد النهايات:

$\lim_{x \rightarrow 0} [e^{2x} + 2x]^{\frac{1}{x}}$	$\lim_{x \rightarrow 0} [e^x - 2x]^{\frac{1}{x}}$	$\lim_{x \rightarrow \infty} \left[\frac{x + 2}{x - 1} \right]^x$
$\lim_{x \rightarrow \infty} \left[\frac{2x + 3}{2x - 1} \right]^x$	$\lim_{x \rightarrow \infty} \left[\frac{x^2 + 1}{x + 2} \right]^{\frac{1}{x}}$	$\lim_{x \rightarrow 0} [e^x + x]^{\frac{1}{x}}$
$\lim_{x \rightarrow \infty} [x]^{\frac{1}{x}}$	$\lim_{x \rightarrow \infty} \left[\frac{x + 1}{x + 2} \right]^x$	

III- Find the exact value of each expression:

أوجد القيمة الدقيقة لكل تعبير

1) $\cos^{-1}\left(\frac{1}{2}\right) + \cos^{-1}\left(-\frac{1}{2}\right) = ?$		3) $\cos^{-1}\left(\frac{\sqrt{3}}{2}\right) + \cos^{-1}\left(-\frac{\sqrt{3}}{2}\right) = ?$	
2) $\cos^{-1}\left(\frac{\sqrt{2}}{2}\right) + \sin^{-1}\left(\frac{\sqrt{2}}{2}\right) = ?$		4) $\cos^{-1}\left(\frac{\sqrt{3}}{2}\right) + \sin^{-1}\left(\frac{\sqrt{3}}{2}\right) = ?$	
5) $\sin^2\left(\frac{\pi}{12}\right) = ?$	6) $\tan(0) = ?$	7) $\tan^{-1}(-\infty) = ?$	
8) $\cos^2\left(\frac{\pi}{8}\right) = ?$	9) $\tan^{-1}(\infty) = ?$		

IV- For what value of a is

من أجل أي قيمة لـ a يكون

1) $f(x) = \begin{cases} -2x & \text{if } x \leq 2 \\ -2ax + 4 & \text{if } x > 2 \end{cases}$	4) $f(x) = \begin{cases} a^2x + 6 & \text{if } x \leq -2 \\ x + 5 & \text{if } x > -2 \end{cases}$
2) $f(x) = \begin{cases} a^2x + 2 & \text{if } x \leq 1 \\ ax + 4 & \text{if } x > 1 \end{cases}$	5) $f(x) = \begin{cases} 2ax + 1 & \text{if } x \leq 2 \\ x + 5 & \text{if } x > 2 \end{cases}$
3) $f(x) = \begin{cases} ax - 2 & \text{if } x \leq 2 \\ -2ax + 4 & \text{if } x > 2 \end{cases}$	6) $f(x) = \begin{cases} 3 + a^2x & \text{if } x \leq 1 \\ -2ax + 1 & \text{if } x > 1 \end{cases}$
Continuous at every x	
مستمرًا عند كل x.	

V- Find the all asymptotes of the curve:		
أوجد جميع الخطوط المقاربة للمنحنى:		
$f(x) = \frac{x^2 - 2}{x + 2}$	$f(x) = \frac{3 - x}{2 - x}$	$f(x) = \frac{2 - x^2}{x + 1}$
$f(x) = \frac{2x + 2}{x - 1}$	$f(x) = \frac{2x^2 + 3}{x - 1}$	

VI- Use logarithmic differentiation to find $y' = \frac{dy}{dx}$:		
استخدم تفاضل اللوغاريتمي لإيجاد $y' = \frac{dy}{dx}$		
$y = \sqrt{3x^2 + 1}$	$y = \frac{(x - 4)^3 \cdot \sqrt{x^3 + 5}}{\sin^{-1}(x)}$	$y = \frac{\tan(x) \cdot \sqrt{x^2 + 1}}{\sqrt[3]{x^3 + 1}}$
$y = \tan(x) \cdot \sqrt{x^2 + 1}$	$y = \frac{(x - 1)^2 \cdot \sqrt{x^2 + 5}}{\tan(x)}$	
$y = \frac{(2x + 1)^3 \cdot \sqrt[3]{x^2 + 1}}{\sec(x)}$	$y = \frac{(x^2 + 2x)^3 \cdot \sqrt[3]{x + 1}}{\tan^{-1}(2x)}$	

VII- Find the first and second derivative of the functions:

أوجد المشتق الأول والثاني للدوال:

- 1) $y = e^{2x} \sin(3x)$
- 2) $y = e^{-3x} \cos(5x)$
- 3) $y = 2x \cos(3x) + e^{2x} \tan(x)$
- 4) $y = \frac{x^2 - 1}{x^2 + x - 3}$
- 5) $y = \sec(x) + x^2 e^{2x}$
- 6) $y = 2\sqrt{x} \cos(x) + \ln(x^2)$

If $y = e^{2x} \cos(\ln(x)) + \tan^{-1}(\sqrt{x})$ find y' .**VIII- Use implicit differentiation to find y' :**- استخدم التفاضل الضمني لإيجاد y' :

$y^3 - 2x + y = 2$	$\ln(x^2 y^2) = e^{(x^2 + y^2)}$	$e^{2x} = \sin(x + 3y)$
$y^2 - 2x - 1 = 2y$	$\ln(xy) = e^{(x+y)}$	$3\sqrt{y} = x - y$
$2y^2 - 4xy = x - 2$	$y^2 - x^2 - \sin(xy) = 0$	

IX- Find the normal and tangent line to curve:

أوجد الخط العمودي والمماس للمنحنى:

- 1) $x^2 + y^2 - \cos(xy) = 2$ at the point $p(0, 1)$.
- 2) $x^2 + y^2 + 3xy = 1$ at the point $p(0, 2)$.
- 3) $y^2 - 2x + 4y - 1 = 4$ at the point $p(0, 1)$.
- 4) $x^2 + y^2 + \sin(xy) = 1$ at the point $p(0, 1)$.
- 5) $x \sin(2y) = y \cos(2x)$ at the point $p(\frac{\pi}{4}, \frac{\pi}{2})$.
- 6) $x^2 - \sqrt{3}xy + 2y^2 = 5$ at the point $p(\sqrt{3}, 2)$.
- 7) $x^2 \cos^2(y) - \sin(y) = 0$ at the point $p(0, \pi)$.

أو:

Verify that the given point is on the curve and find the lines that are (a) tangent and (b) normal to the curve at the given point.

تحقق من أن النقطة المعطاة تقع على المنحنى وأوجد المستقيمات التي تكون: (أ) مماسة و (ب) عمودية على المنحنى عند النقطة المعطاة.